

THE STRUCTURE OF MATHEMATICAL REALITY

Phase V — Chapter 16

Category Theory: The Mathematics of Mathematics

Assignment

Self-Study Curriculum

Return to your diagram of the mathematical fields (revised in Chapter 3). Now add the following layer:

For each bridge between fields that you identified, describe it as a functor: what objects are mapped to what objects? What morphisms are mapped to what morphisms?

You do not need to be technically precise. The goal is to see that the connections between fields are not ad hoc, they are systematic, structure-preserving translations.

Write a half-page reflection: does category theory resolve any of the “connective tissue” problems you started with? Does it create new questions?